

4. Suppose ξ is a root of order 2 of $f(x)$, meaning $f(\xi) = 0$, $f'(\xi) = 0$ and $f''(\xi) \neq 0$.

(a) Show that in this case Newton's method no longer converges quadratically. Do so by showing that the method now converges linearly.

(b) We can modify Newton's method in this case: Show that the iteration

$$x_{n+1} = x_n - 2 \frac{f(x_n)}{f'(x_n)}$$

does converge quadratically (or faster) when ξ is a root of order 2 of $f(x)$.

Throughout this problem you may assume that $f(x)$ has as many continuous derivatives as you need, for instance that $f'''(x)$ is continuous in a neighborhood of ξ .

Recommendation: Use a Taylor series for $h(x) = f(x)/f'(x)$, similar to what we did in lecture when examining convergence of Newton's method. You do not need anything more complicated. However, you will find it useful to remember L'Hôpital's rule from calculus when computing $h(\xi)$ and $h'(\xi)$. Do not worry about the exact value of $h''(\xi)$.

$$f(\xi) = 0 \quad f'(\xi) = 0 \quad f''(\xi) \neq 0.$$

for newton methods:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = g(x)$$

$$g(x) = g(\xi) + g'(\xi)(x - \xi) + \frac{1}{2} g''(\xi)(x - \xi)^2$$

$$\frac{x_{n+1} - \xi}{x_n - \xi} = g'(\xi) + \frac{1}{2} g''(\xi)(x_n - \xi)$$

when $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} \frac{|x_{n+1} - \xi|}{|x_n - \xi|} \leq |g'(\xi)|$$

$$f(x) = (x - \xi)^2 h(x) \quad h(\xi) \neq 0 \Rightarrow f'(\xi) \neq 0.$$

$$f'(x) = 2(x - \xi)h(x) + (x - \xi)^2 h'(x).$$

$$g(x) = x - \frac{(x - \xi)^2 h(x)}{2(x - \xi)h(x) + (x - \xi)^2 h'(x)} = x - \frac{(x - \xi)h(x)}{2h(x) + (x - \xi)h'(x)}$$

$$g'(x) = 1 - \frac{(h(x) + (x - \xi)h'(x))(2h(x) + (x - \xi)h'(x)) - (x - \xi)h(x)(2h'(x) + (x - \xi)h''(x) + h'(x)))}{(2h(x) + (x - \xi)h'(x))^2}$$

$$\lim_{n \rightarrow \infty} g'(x) = 1 - \frac{2(h(x))^2}{4(h(x))^2} \Rightarrow g'(\frac{1}{2}) = \frac{1}{2}.$$

$$\lim_{n \rightarrow \infty} \frac{|x_{n+1} - \frac{1}{2}|}{|x_n - \frac{1}{2}|} \leq g'(\frac{1}{2}) = \frac{1}{2}$$

$\Rightarrow$ which means the function converges linearly and does not converge quadratically.

$$b). \quad x_{n+1} = x_n - 2 \frac{f(x)}{f'(x)}$$

$$g(x) = x_n - 2 \frac{f(x)}{f'(x)}$$

$$\lim_{n \rightarrow \infty} \frac{g(x)}{g'(x)} = 1 - 2 \frac{2(h(x))^2}{4(h(x))^2} = 0$$

$$g(x) = g(\frac{1}{2}) + g'(\frac{1}{2})(x - \frac{1}{2}) + \frac{1}{2} g''(\frac{1}{2})(x - \frac{1}{2})^2 + \frac{1}{6} g'''(\frac{1}{2})(x - \frac{1}{2})^3$$

$$x_{n+1} = \frac{1}{2} + g'(\frac{1}{2})(x_n - \frac{1}{2}) + \frac{1}{2} g''(\frac{1}{2})(x_n - \frac{1}{2})^2 + \frac{1}{6} g'''(\frac{1}{2})(x_n - \frac{1}{2})^3$$

$$x_{n+1} - \frac{1}{2} = \frac{1}{2} g''(\frac{1}{2})(x_n - \frac{1}{2})^2 + \frac{1}{6} g'''(\frac{1}{2})(x_n - \frac{1}{2})^3$$

$$\frac{x_{n+1} - \frac{1}{2}}{(x_n - \frac{1}{2})^2} = \frac{1}{2} g''(\frac{1}{2}) + \frac{1}{6} g'''(\frac{1}{2})(x_n - \frac{1}{2})$$

$$\text{for } n \rightarrow \infty \quad \lim_{n \rightarrow \infty} \frac{|x_{n+1} - \frac{1}{2}|}{|x_n - \frac{1}{2}|^2} \leq \frac{1}{2} |g''(\frac{1}{2})|$$

Since $g''(\frac{1}{2}) \neq 0$, which means the function converge quadratically when we use $x_{n+1} = x_n - 2 \frac{f(x)}{f'(x)}$.

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function [x, xhist] = muller(f, x0, x1, x2, tol, maxit)

    if nargin < 6, maxit = 100; end
    if nargin < 5, tol = 1e-12; end

    xhist = zeros(maxit+2,1);
    xhist(1:3) = [x0; x1; x2];

    for k = 1:maxit
        f0 = f(x0); f1 = f(x1); f2 = f(x2);

        h0 = x1 - x0;
        h1 = x2 - x1;

        d0 = (f1 - f0) / h0;
        d1 = (f2 - f1) / h1;

        a = (d1 - d0) / (h1 + h0);
        b = a*h1 + d1;
        c = f2;

        disc = sqrt(b.^2 - 4*a*c);
        if abs(b + disc) > abs(b - disc)
            denom = b + disc;
        else
            denom = b - disc;
        end

        if abs(denom) == 0
            x3 = x2;
        else
            x3 = x2 + (-2*c) / denom;
        end

        xhist(k+3) = x3;

        if abs(x3 - x2) <= tol*(1 + abs(x3)) || abs(f(x3)) <= tol
            xhist = xhist(1:k+3);
            x = x3;
            return;
        end

        x0 = x1; x1 = x2; x2 = x3;
    end

    x = x2;
    xhist = xhist(1:maxit+2);
end

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f = @(x) x.^3 + x + 1;
[root1,it1] = muller(f,-1.0,0.0,1.0);

p = [1 0 1 1];
[q,~] = deconv(p,[1 -root1]);

a = q(1); b = q(2); c = q(3);
disc = b^2 - 4*a*c;
root2 = (-b + sqrt(disc))/(2*a);
root3 = (-b - sqrt(disc))/(2*a);
fprintf('root1 ≈ %.5e%.5ei (iters=%d)\n', real(root1), imag(root1), it1);

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root1 ≈ -6.82328e-01+0.00000e+00i (iters=-1)
root1 ≈ 0.00000e+00+1.00000e+00i (iters=-5.00000e-01)
root1 ≈ -1.00000e+00-6.45751e-01i (iters=-6.814658e-01)
root1 ≈ -6.82324e-01-6.82328e-01i (iters=-6.823278e-01)

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fprintf('root2 ≈ %.5e%.5ei\n', real(root2), imag(root2));

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root2 ≈ 3.41164e-01+1.16154e+00i

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fprintf('root3 ≈ %.5e%.5ei\n', real(root3), imag(root3));

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root3 ≈ 3.41164e-01-1.16154e+00i

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